Capstone Project Monthly Progress Report

Month: February 2026

Project Title: Biometric Template Generation with Cancelable Biometrics

Project ID: UROP_2022 _24250_01

Student Name: Shanmuka Srinivasa Siva Sai M
Student Roll Number: AP22110010438
Total Number of Students in Team: 4

Details of Team Members:
Sai Pranav SVSS (Roll No. AP22110010441)
Shanmuka Srinivasa Siva Sai M (Roll No. AP22110010438)
Sai Naidu P (Roll No. AP22110010401)
Charith J (Roll No. AP22110010444)

Supervisor: Dr. Dilip Kumar V

Department of Computer Science and Engineering
SRM University AP

---

1 Project Overview

This project focuses on biometric template generation using deep learning techniques with privacy-preserving cancelable biometrics. The system aims to develop a secure biometric authentication framework that protects user privacy while maintaining high recognition accuracy across multiple fingerprint datasets.

2 Work Completed During Last 1 Month

- Designed and implemented a deep neural network architecture (ResNet50 + ArcFace -based model) for fingerprint feature extraction
- Developed biometric cryptosystem modules including cancelable transforms and key management
- Addressed reviewer comments for conference paper submission
- Trained models on both FVC2000 and FVC2006 fingerprint datasets
- Generated embeddings and performed initial evaluation metrics

3 Work Planned for this Month

- Complete model evaluation on test sets

- Finalize cancelable biometric benchmarking
- Submit revised conference paper
- Prepare documentation and final report

4 Progress Status

- Overall completion: 50%
- Model training completed on both datasets
- Biometric cryptosystem modules integrated
- Paper revisions in progress

5 Challenges/Issues Faced

- Initial model convergence issues - resolved by adjusting learning rate and margin parameters
- High-dimensional embeddings required adaptation of traditional fuzzy commitment schemes
- Cross-dataset evaluation showed domain shift challenges between FVC2000 and FVC2006

6 Learning Outcomes

- Gained expertise in ResNet50 + ArcFace and metric learning for biometric recognition
- Learned cryptographic primitives including PBKDF2, AES for template protection
- Understood cancelable biometric design principles and privacy preservation
- Improved academic writing and peer review response skills

7 Plan for Next Month

- Complete all model evaluations and generate comprehensive results
- Finalize and submit revised conference paper
- Complete benchmarking analysis
- Prepare final project documentation

8 Student Individual Contribution

Trained the deep learning model on FVC2006 fingerprint dataset using transfer learning from FVC2000. Successfully adapted the model to handle different image characteristics, performed cross-dataset evaluation between FVC2000 and FVC2006, and achieved improved recognition performance on the FVC2006 benchmark.

9. Supervisor's Observation

Student Completed: 80 % of the planned/assigned task during the last month

Comments:

→ Need to complete implementation of cryptosystem

Number of times students meet with supervisor (Online or Offline): 4

Satisfied with the Student's Work: ✓ Satisfied Not Satisfied

Supervisor Signature: V. Dilip Kumar

Date: 25/02/2026